CONCEPTUAL OVERVIEW

The idea: decode all variables and the relationships between them

Measure current values, assignment history, data dependencies and algebraic relations from model representations.

Program with updates

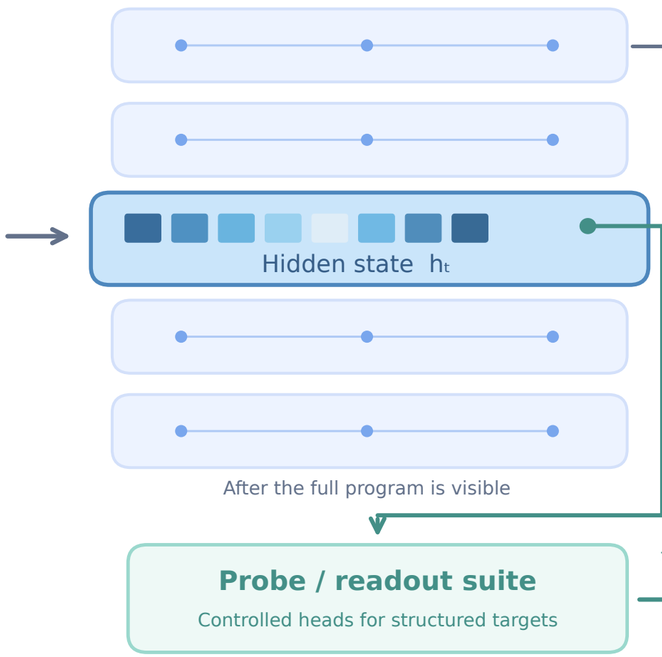
```
a = 2
b = 7
saved = a
a = b - a
out = a + saved
print(out)
```

Python output: 7

Four variables; five assignment events.

Values alone do not identify their owners.

Language model



Probe / readout suite

Controlled heads for structured targets

targets

Generated answer — One response, scored separately ?

Structured decoding targets

Illustrative oracle labels shown below

All current bindings

a@2	5
b@1	7
saved@1	2
out@1	7

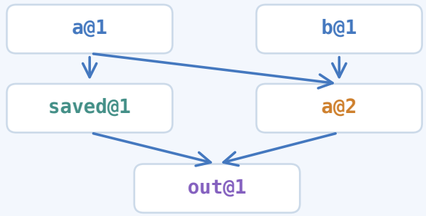
Current + previous versions

a@1 = 2 → a@2 = 5

previous — current

saved keeps a@1's value

Data dependencies



Algebraic relationships

$a < b$

`out = a + saved`

$b = out$

Do the right bindings and relationships remain accessible—and does the model use them?

Decode broadly → test selective causal effects → predict failures on unfamiliar programs

Conceptual framework. Values and edges are program ground truth, not measured probe predictions or evidence of a complete neural state store.